

Feedback Amplifiers (Prabin)

1) Importance of positive feedback on oscillation.

So A transistor amplifier with proper positive feedback can act as an oscillator.

In positive feedback, signal is in phase with the original input.

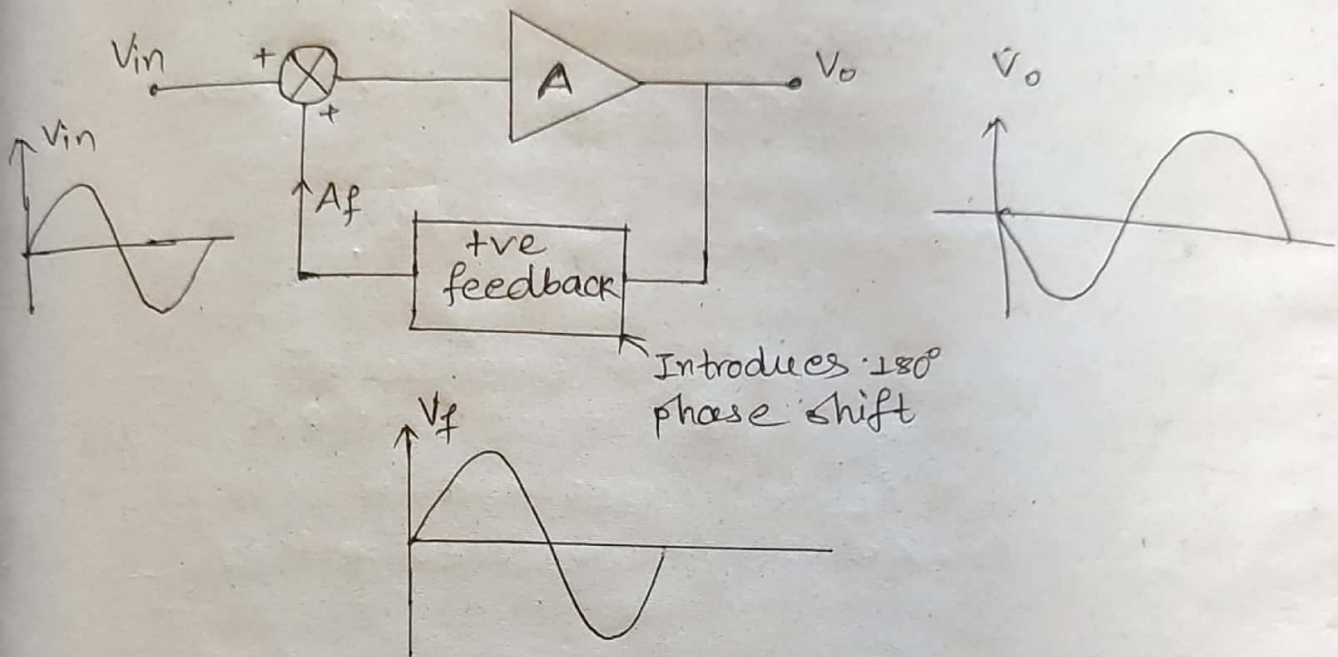


fig. positive feedback

The following points may be noted carefully :-

- i) The transistor amplifier with proper positive feedback will work as an oscillator.
- ii) In order to get continuous undamped output from the circuit from the expression of closed loop gain with positive feedback,

$$A_f = \frac{A}{1 - AB}$$

where,

A_f = positive feedback closed loop gain

A = open loop gain of an amplifier

B = feedback ratio

The following condition must be satisfied.
i.e. $AB = 1$ & $A_f = \infty$

Even a small noise signal can provide a measurable output voltage and the circuit acts as an oscillator.

- 5.b. A negative feedback of 0.2% is applied to an amplifier with gain of 60 dB. Calculate the percentage change in the overall voltage gain of the feedback amplifier if the internal amplifier is subjected to gain of reduction of 15%.

sol

Given:

Feedback ratio, $\beta = 0.2\% = 0.002$

open loop gain, $A_{dB} = 60 \text{ dB}$

change in open loop gain, $\frac{dA}{A} = 15\%$

change in feedback gain, $\frac{dA_f}{A_f} = ?$

we know

$$A_{dB} = 20 \log A$$

$$\Rightarrow 60 = 20 \log A$$

$$\Rightarrow \frac{60}{20} = \log A$$

$$\Rightarrow A = 10^3 = 1000$$

Now,

$$\frac{dA_f}{A_f} = \frac{1}{(1 + \beta A)} \times \frac{dA}{A}$$

$$= \frac{1}{(1 + 0.002 \times 1000)} \times 15\%$$

$$= \frac{15}{3}\%$$

$$\therefore \frac{dA_f}{A_f} = 5\%$$

Thus the percentage change in the overall voltage gain is 5%.

5. b. What are positive and negative feedback? Negative feedback extends the bandwidth. Explain it with necessary derivations.

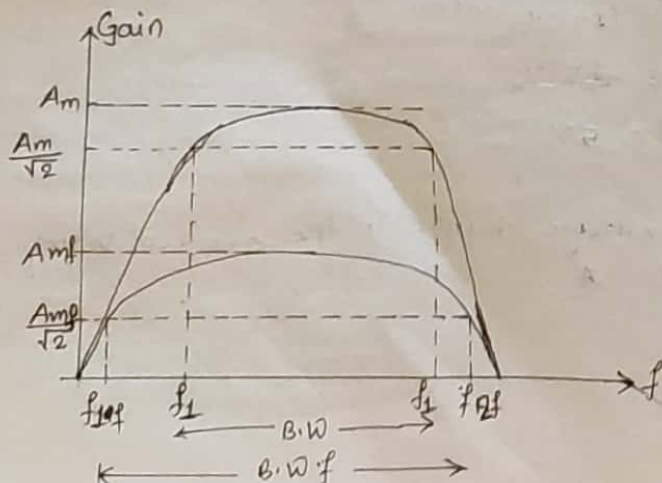
So? Positive feedback is the feedback which is used to increase the input signal. It is also called direct or regenerative feedback.

Negative feedback is the feedback which is applied to reduce the input signal. It is also called inverse or degenerative feedback.

Negative feedback extends the bandwidth!

If f_L and f_H are lower and upper cut off frequency of an amplifier without feedback then the values of f_L decreases and f_H increases by factor of $(1 + \beta A_m)$, if negative feedback is used.

The plot of gain and frequency of an amplifier is shown below:



Here, without feedback,

$$BW = f_2 - f_1$$

where f_1 = lower cut off frequency

f_2 = higher/upper cut off frequency

After negative feedback,

$$BW_f = f_{2f} - f_{1f}$$

$$\Rightarrow BW_f = f_2(1 + \beta A_m) - \frac{f_1}{(1 + \beta A_m)}$$

$$\Rightarrow BW_f = f_2(1 + \beta A_m) \left(\because \frac{f_1}{(1 + \beta A_m)} \gg \frac{f_1}{(1 + \beta A_m)} \right)$$

$$\Rightarrow BW_f = BW * (1 + \beta A_m) \left(\because BW = f_2 - f_1 \approx f_2 \right)$$

$$\therefore BW_f = BW(1 + \beta A)$$

Thus the negative feedback extends bandwidth

6.b. Calculate overall gain of a negative feedback amplifier. Also list out advantages of negative feedback.

Sol Negative feedback is the feedback which is applied to reduce input signal is called negative feedback. It is also called inverse or degenerative feedback.

The feedback amplifier introduces a phase shift of 180° into the circuit, while the feedback network is so designed that it reduces no phase shift. Thus it results that the feedback voltage V_f is 180° out of phase with the input signal V_{in} .

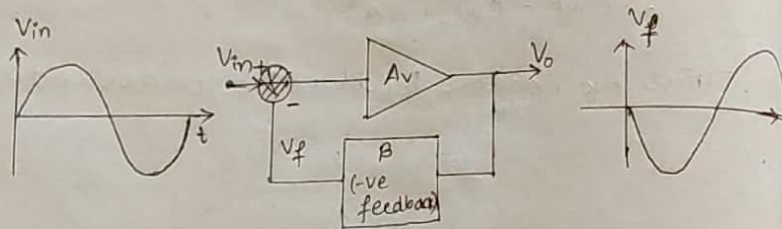


fig. negative feedback

we have,

$$V_o = A \times (V_{in} - V_f) \quad \text{--- (1)}$$

$$\text{Also, } V_f = \beta V_o \quad \text{--- (2)}$$

From (1) & (2), we get;

$$V_o = AV_{in} - A\beta V_o$$

$$\Rightarrow V_o + A\beta V_o = AV_{in}$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{A}{1 + \beta A}$$

$$\therefore A_f = \frac{V_o}{V_{in}} = \frac{A}{1 + \beta A}$$

This is the overall gain of a negative feedback amplifier.

The negative feedback reduces the gain of amplifier. However, it has many advantages so that it is frequently used in amplifier circuits, which are as given below:-

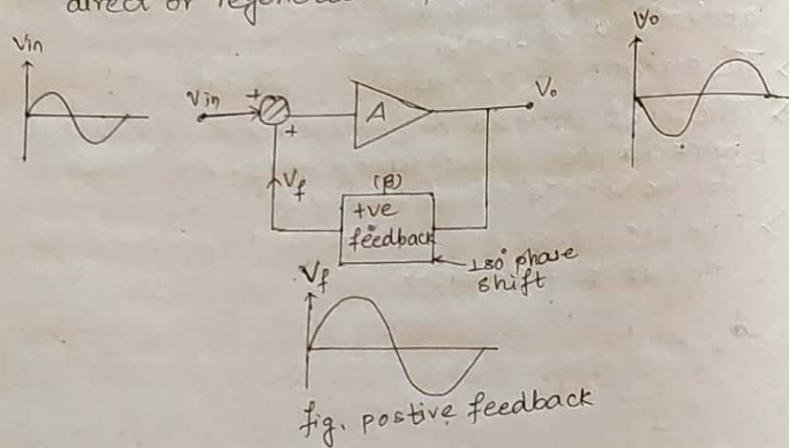
- i) It increases the gain stability
- ii) The non-linear distortion is reduced by a factor of $(1 + A\beta)$.
- iii) The bandwidth of the amplifier is extended by approximately $(1 + A\beta)$ times.
- iv) The signal to noise ratio is improved.
- v) The input impedance is increased and output impedance is decreased by $(1 + A\beta)$ times for the voltage series feedback configuration.
- vi) The frequency distortion is reduced with the negative feedback.

5. b. Derive a closed loop gain expression for feedback amplifier and explain the characteristics of negative feedback amplifier.

Sol There are two types of feedback amplifiers. They are

i) Positive feedback Amplifier

Positive feedback is the feedback signal which is applied to increase the input signal. It is also called direct or regenerative feedback.



In above fig, A is open loop gain and β is feedback ratio, we have

$$V_o = (V_{in} + V_f)A \quad (1)$$

$$\text{Also, } V_f = \beta V_o \quad (2)$$

From (1) & (2),

$$V_o = (V_{in} + \beta V_o)A$$

$$\Rightarrow V_o - \beta A V_o = V_{in} A$$

$$\Rightarrow V_o (1 - \beta A) = V_{in} A$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{A}{1 - \beta A}$$

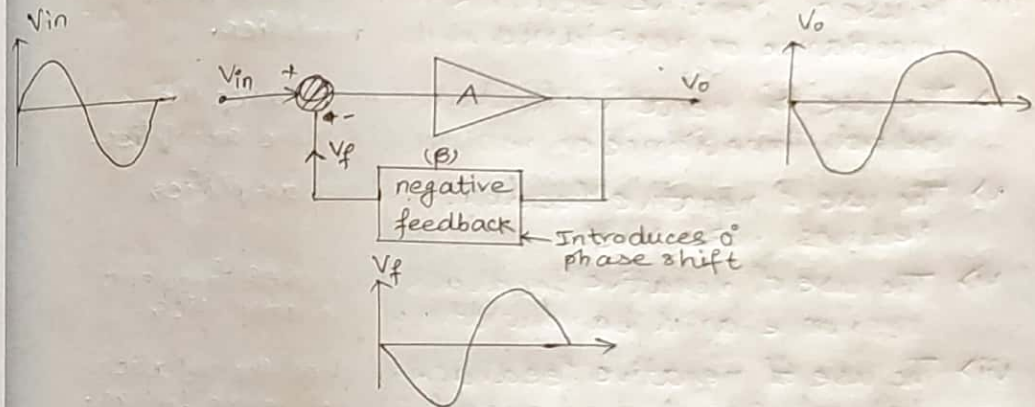
$$\Rightarrow \therefore V_f = \frac{A}{1 - \beta A}$$

$$\therefore V_f = \frac{A}{1 - \beta A}$$

This is closed loop gain for positive feedback amplifier.

ii) Negative feedback Amplifier

Negative feedback is the feedback signal which is applied to decrease the input signal. It is also called inverse or degenerative feedback.



In the above figure, A is open loop gain and β is feedback ratio, we have

$$V_o = (V_{in} - V_f)A \quad (1)$$

$$\text{Also, } V_f = \beta V_o \quad (2)$$

From (1) & (2), we get:

$$V_o = (V_{in} - \beta V_o)A$$

$$\Rightarrow V_o = A V_{in} - \beta A V_o$$

$$\Rightarrow V_o + \beta A V_o = A V_{in}$$

$$\Rightarrow V_o (1 + \beta A) = A V_{in}$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{A}{1 + \beta A}$$

$$\therefore V_f = \frac{A}{1 + \beta A}$$

$$\therefore V_f = \frac{A}{1 + \beta A}$$

This is closed loop gain for negative feedback amplifier.

The characteristics of negative feedback amplifiers :-

- i> The negative feedback stabilizes the gain of the amplifier by reducing the dependence of amplifier gain on various transistor parameters or variation in the supply voltage.
- ii> The negative feedback also reduces the non-linear distortion level in the large signal amplifiers.
- iii> The use of negative feedback increases the input impedance or resistance of the amplifier.
- iv> The use of negative feedback decreases the output impedance or resistance of the amplifier.
- v> The use of negative feedback ~~reduces~~ amplifier reduces noise level.
- vi> The use of negative feedback improves frequency response and bandwidth of the amplifier.
- vii> The use of negative feedback makes the relation between input and output signal more linear.

5. b. An amplifier has an input 20 mV and a gain of 150 without the feedback. The distortion produced at the output of the amplifier is 10%. It is desired to reduce the distortion to 2% by using negative feedback. Calculate the gain and the output voltage with the feedback.

sol Given input voltage, $V_{in} = 20 \text{ mV}$

Gain, $A = 150$

$$\frac{dA}{A} = 10\%$$

$$\frac{dA_f}{A_f} = 2\%$$

$$A_f = ?$$

$$V_o = ?$$

We know that,

$$\frac{dA_f}{A_f} = \frac{dA}{A} \times \frac{1}{1 + AB}$$

$$\Rightarrow 2 = 10 \times \frac{1}{1 + 150 \times B}$$

$$\Rightarrow \frac{1}{5} = \frac{1}{1 + 150B}$$

$$\Rightarrow 5 = 1 + 150B$$

$$\Rightarrow 5 - 1 = 150B$$

$$\Rightarrow \frac{4}{150} = B$$

$$\therefore B = 0.0267$$

Now,

$$\text{Gain, with feed back, } A_f = \frac{A}{1 + AB}$$

$$= \frac{150}{1 + 150 \times \frac{2}{75}}$$

$$\therefore A_f = 30$$

we know,

$$A_f = \frac{V_o}{V_{in}}$$

$$\Rightarrow V_o = A_f \cdot V_{in}$$

$$\Rightarrow V_o = 80 \times 20 \times 10^{-3}$$

$$\therefore V_o = 0.6 \text{ V}$$

6. a. Explain the effect of negative feedback configuration in amplifier configuration characteristics

6.b Explain the advantages and disadvantages of negative feedback. A negative feedback of 0.2% is applied to an amplifier gain of 60 dB. Calculate the percentage change in the overall gain of the feedback amplifier if the internal amplifier is subjected to gain reduction of 15%.

Sol:

The advantages of negative feedback are:-

- i) It increases the gain stability.
- ii) The non-linear distortion is reduced by a factor of $(1+AB)$.
- iii) The bandwidth of the amplifier is extended by approximately $(1+AB)$ times.
- iv) The signal to noise ratio is improved.
- v) The input impedance is increased and output impedance is decreased by $(1+AB)$ times for the voltage series feedback configuration.
- vi) The frequency distortion is reduced with the negative feedback.

The disadvantages of negative feedback are:-

- i) Its reduction in gain.
- ii) It increases output resistance in the case of shunt and current series feedback amplifiers.

Numerical:

$$\text{Open loop gain (A)} = 60 \text{ dB}$$

$$\Rightarrow 20 \log_{10} A = 60 \Rightarrow \log_{10} A = 3$$

$$\Rightarrow A = 10^3 = 1000$$

$$\therefore \frac{dA}{A} = 15\%$$

$$\therefore \beta = 0.2\% = 0.002$$

we know,

$$\frac{dA_f}{A_f} = \frac{dA}{A} \times \frac{1}{1 + A\beta}$$

$$\Rightarrow \frac{15}{1000} \frac{dA_f}{A_f} = \frac{15}{100} \times \frac{1}{(1 + 1000 \times 0.002)}$$

$$\therefore \frac{dA_f}{A_f} = \frac{1}{20} = 5\%$$

Thus the % change in the overall gain of the feedback amplifier is 5%. Ans

6.b. Design an operational amplifier circuit that produce an output $V_o = 8V_1 + 10V_2 + 0.8V_3 + 4V_4$.

sof We have,

$$V_o = 8V_1 + 10V_2 + 0.8V_3 + 4V_4$$
$$\Rightarrow V_o = -[-8V_1 - 10V_2 - 0.8V_3 - 4V_4] \quad \text{--- (1)}$$

we know that the op amp summing amplifier is

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 + \frac{R_f}{R_4} V_4 \right] \quad \text{--- (2)}$$

comparing (1) & (2), we get;

$$\frac{R_f}{R_1} V_1 = -8V_1 \quad \therefore \frac{R_f}{R_1} = -8 \quad \text{--- (3)}$$

$$\frac{R_f}{R_2} V_2 = -10V_2 \quad \therefore \frac{R_f}{R_2} = -10 \quad \text{--- (4)}$$

$$\frac{R_f}{R_3} V_3 = -0.8V_3 \quad \therefore \frac{R_f}{R_3} = -0.8 \quad \text{--- (5)}$$

$$\frac{R_f}{R_4} V_4 = -4V_4 \quad \therefore \frac{R_f}{R_4} = -4 \quad \text{--- (6)}$$

Suppose $R_f = 20 \text{ K}\Omega$. so

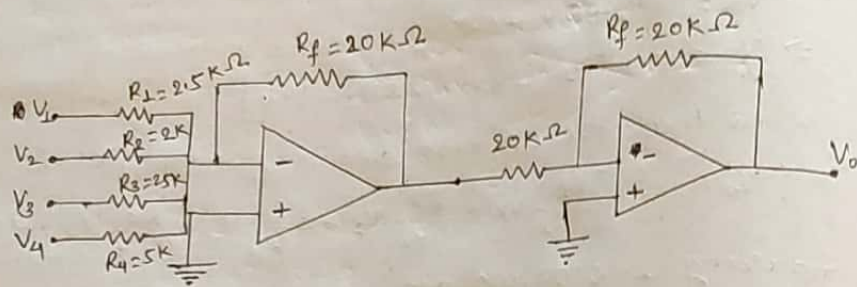
$$\text{from (3), } \frac{R_f}{R_1} = -8 \Rightarrow R_1 = \frac{20}{-8} = -2.5 \text{ K}\Omega$$

$$\text{from (4), } \frac{R_f}{R_2} = -10 \Rightarrow R_2 = \frac{20}{-10} = -2 \text{ K}\Omega$$

$$\text{from (5), } \frac{R_f}{R_3} = -0.8 \Rightarrow R_3 = \frac{20}{-0.8} = -25 \text{ K}\Omega$$

$$\text{from (6), } \frac{R_f}{R_4} = -4 \Rightarrow R_4 = \frac{20}{-4} = -5 \text{ K}\Omega$$

Thus the required op amp circuit that produce an output
 $V_o = 8V_1 + 10V_2 + 0.8V_3 + 4V_4$ is:-



(2021-Fall)

6.a. Calculate the gain of inverting amplifier using an operational amplifier. Show how it can be used to build an adder circuit. Explain with necessary diagram and expression.

sol. The circuit diagram for inverting op amp is as shown below:-

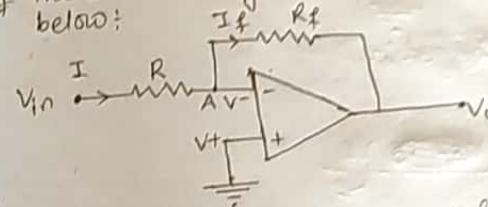


fig. inverting amplifier

Suppose ▢, V_{in} = input voltage
 R = input resistance
 R_f = feedback resistance
 V^+ = voltage at positive terminal
 V^- = voltage at negative terminal
 V_o = output voltage

Applying KCL at node A, we get:-
 $I = I_f$

$$\Rightarrow \frac{V_{in} - V_A}{R} = \frac{V_A - V_o}{R_f}$$

$$\Rightarrow \frac{V_{in}}{R} = -\frac{V_o}{R_f} \quad (\because V_A = 0, \text{ virtual ground})$$

$$\Rightarrow -\frac{R_f}{R} = \frac{V_o}{V_{in}}$$

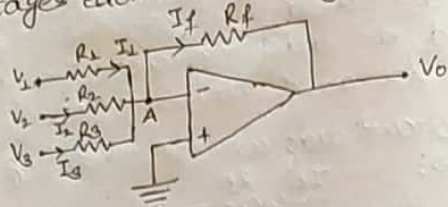
$$\Rightarrow \therefore A_f = -\frac{R_f}{R} \quad (\because A_f = V_o/V_{in})$$

$$\therefore A_f = -\frac{R_f}{R}$$

This is required gain of inverting amplifier using an op amp.

Adder or Summer Amplifier:

The function of summer or adder is to provide an output which is proportional to the algebraic sum of two or more input voltages each multiplied by a constant gain factor.



Applying KCL at node A, we get:-

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} + \frac{V_3 - V_A}{R_3} = \frac{V_A - V_0}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{V_A - V_0}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_0}{R_f}$$

$$\Rightarrow \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 = -V_0$$

$$\Rightarrow V_0 = -\frac{R_f}{1} \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Suppose $R_1 = R_2 = R_3 = R$, so

$$V_0 = -\frac{R_f}{1} \left(\frac{V_1}{R} + \frac{V_2}{R} + \frac{V_3}{R} \right)$$

$$\therefore V_0 = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

This is the output voltage of adder amplifier which is proportional to the algebraic sum of the input voltages

6. b. Explain Barkhausen criteria of oscillation. Design Wein-bridge oscillator to operate at a frequency of 256 KHz.

The closed loop gain of positive feedback amplifier is $A_f = \frac{V_0}{V_{in}} = \frac{A}{1 - AB}$

where, A = gain of an amplifier
 B = feedback ratio (factor)

when $AB = 1$,

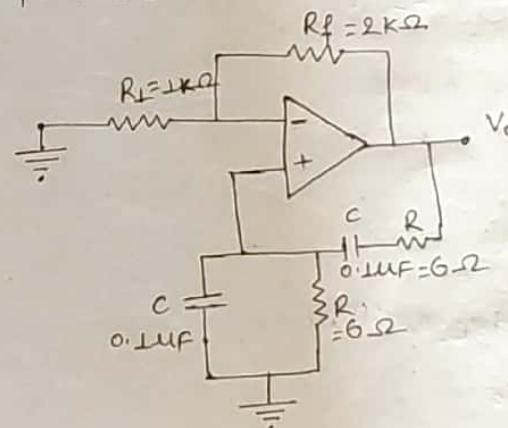
$$A_f = \frac{A}{1-1} = \frac{A}{0} = \infty$$

This means output may appear with very small input ideally no input can also produce an output.

So the Barkhausen criteria for oscillations are as follows:

- i) Loop gain must be unity, i.e. $AB = 1$.
- ii) There must be positive feedback i.e. input and feedback should be in same phase.

The Wein-bridge oscillator to operate at a frequency of 256 KHz is.



Given: $f = 256 \text{ KHz}$

Let us assume, $C = 0.1 \mu\text{F} = 0.1 \times 10^{-6} \text{ F}$

By Wein bridge oscillator frequency,

$$f = \frac{1}{2\pi RC}$$

$$\Rightarrow R = \frac{1}{2\pi fC}$$

$$\Rightarrow R = \frac{1}{2\pi \times 0.1 \times 10^{-6} \times 256 \times 10^3}$$

$$\therefore R = 6216 \Omega$$

$$\text{i.e. } R \approx 6 \Omega$$

As we know,

$$\therefore B = \frac{1}{3}$$

$$\therefore AB = 1 \Rightarrow A = \frac{1}{B} \Rightarrow A = \frac{1}{1/3}$$

$$\therefore A = 3$$

For non-inverting amplifier,

$$A = 1 + \frac{R_f}{R_1} \Rightarrow 3 = 1 + \frac{R_f}{R_1}$$

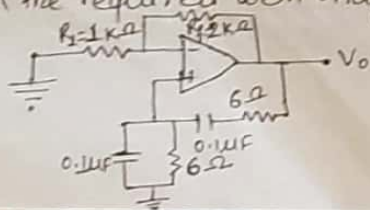
$$\Rightarrow 3 - 1 = \frac{R_f}{R_1}$$

$$\Rightarrow R_1 = \frac{R_f}{2}$$

$$\text{Let } R_1 = 1 \text{ k}\Omega$$

$$\therefore R_f = 2 \text{ k}\Omega$$

Then the required Wein bridge oscillator is



(2020 - Fall)

6.a. Find out the output voltage expression for differentiator and integrator circuit. Design an adder circuit using op. amplifiers to get output expression given by $V_o = 3V_1 - 15V_2 - 120V_3$.

sol. Expression for differentiator:

The function of the differentiator is to provide an output which is proportional to the rate of change of input voltage.

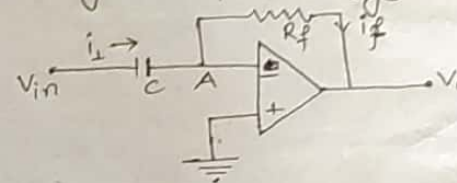


fig. OP-amp as differentiator

As we know,

$$\text{current, } i = \frac{dQ}{dt}$$

$$\text{Also, } Q = CV$$

$$\text{Now, } i_1 = \frac{d(CV)}{dt} = \frac{d(CV_{in})}{dt} = C \frac{dV_{in}}{dt}$$

From figure,

$$i_f = \frac{V_A - V_o}{R_f} = -\frac{V_o}{R_f} \quad [\because V_A = 0, \text{ virtual ground}]$$

Applying KCL at node A, we get;

$$i_1 = i_f \quad (\because \text{incoming current} = \text{outgoing current})$$

$$\Rightarrow C \frac{dV_{in}}{dt} = -\frac{V_o}{R_f}$$

$$\therefore V_o = -R_f C \frac{dV_{in}}{dt}$$

This is the output voltage expression for differentiator.

Output voltage expression for integrator:
The function of the integrator is to provide an output which is proportional to the integral of input voltage.

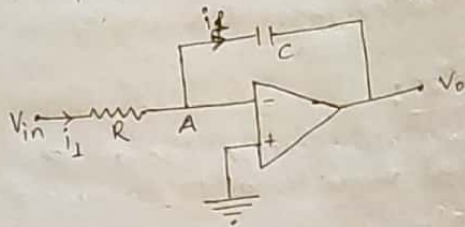


fig. Op amp as integrator

As we know,
current, $i = \frac{dq}{dt} = \frac{d(cV)}{dt} = c \frac{dV}{dt}$
 $\therefore i_f = c \frac{dV_o}{dt} = c \frac{d(V_A - V_o)}{dt} = -c \frac{dV_o}{dt} \quad (\because V_A = 0)$

Also,
 $i_1 = \frac{V_{in} - V_A}{R} = \frac{V_{in}}{R}$

Applying KCL at node A, we get:

$$i_1 = i_f$$

$$\Rightarrow \frac{V_{in}}{R} = -c \frac{dV_o}{dt}$$

$$\Rightarrow -\frac{V_{in}}{Rc} = \frac{dV_o}{dt}$$

$$\therefore V_o = -\int \frac{V_{in}}{Rc} dt$$

$$\therefore V_o = -\frac{1}{Rc} \int V_{in} dt$$

This is the output voltage expression for integrator op-amp.

6. b. What is Barkhausen Criteria for Oscillation? Design a RC phase shift oscillator to operate at a frequency of 5 KHz.

sol. The closed loop gain of positive feedback amplifier is

$$A_f = \frac{V_o}{V_{in}} = \frac{A}{1 - AB}$$

where, A = gain of an amplifier,
B = feedback ratio (factor)

when $AB = 1$.

$$A_f = \frac{A}{1-1} = \frac{A}{0} = \infty$$

This means output may appear with very small input, ideally no input can also produce an output.

So the Barkhausen Criteria for oscillations are as follows:

i) loop gain must be unity i.e. $AB = 1$.

ii) There must be positive feedback i.e. input and feedback should be in same phase.

We have,

frequency, $f = 5 \text{ KHz}$

The frequency of oscillation of RC phase shift oscillator is given by

$$f = \frac{1}{2\sqrt{6} \pi RC} \quad \text{--- (1)}$$

Suppose, $R = 1 \text{ K}\Omega$ from (1),

$$5 \times 10^3 = \frac{1}{2\sqrt{6} \pi \times 10^3 \times C}$$

$$\Rightarrow 5 \times 10^6 \times 2\pi\sqrt{6} = \frac{1}{C}$$

$$\Rightarrow C = \frac{1}{76952.989}$$

$$\therefore C = 12.99 \times 10^{-9} = 13 \text{ nF}$$

Since $R_2 = 29 K\Omega$
 $\Rightarrow R_2 = 29 \times 1 K\Omega$
 $\therefore R_2 = 29 K\Omega$

Thus the required RC phase shift oscillator is shown below:

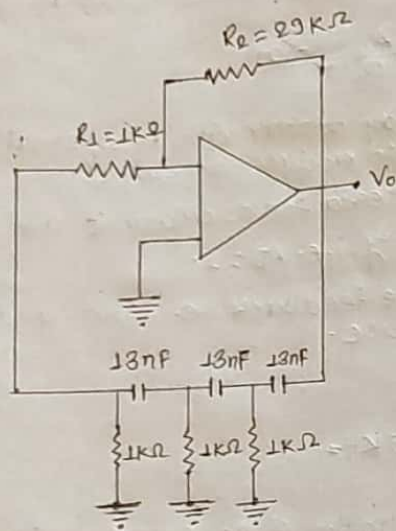


fig. RC phase shift oscillator of frequency 5 KHz

(2019 - Spring)

6.a. Find out the output voltage expression for inverting and summing circuit with circuit diagram and Explain the characteristics of practical operational amplifier.

Q. The function of summing circuit is to provide an output which is proportional to the algebraic sum of two or more input voltages each multiplied by a constant gain factor.

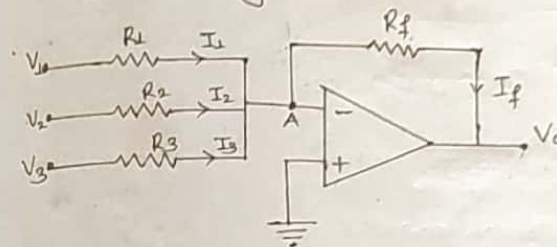


fig. adder op amp

Applying KCL at node A, we get:

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} + \frac{V_3 - V_A}{R_3} = \frac{V_A - V_o}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

Let $R_1 = R_2 = R_3 = R$. So

$$\frac{V_1}{R} + \frac{V_2}{R} + \frac{V_3}{R} = -\frac{V_o}{R_f}$$

$$\Rightarrow -R_f \left(\frac{V_1 + V_2 + V_3}{R} \right) = V_o$$

$$\therefore V_o = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

This is the output voltage expression for summing circuit.

The characteristics of practical operational amplifier are as follows:-

- i) Open loop voltage gain is high.
- ii) Large input impedance.
- iii) Very low output impedance.
- iv) Finite bandwidth.
- v) Small offset voltage.
- vi) Finite CMRR.

6.b. Write the salient features of operational amplifier. Design a summing operational amplifier circuit that will produce an output voltage $V_o = -4V_1 - V_2 + 0.1V_3$. The salient features of operational amplifier are as follows:-

- i) Open loop voltage gain is high.
- ii) Large input impedance.
- iii) Very low output impedance.
- iv) Finite Bandwidth.
- v) Small offset voltage.
- vi) Finite CMRR.

The given output voltage is

$$V_o = -4V_1 - V_2 + 0.1V_3$$

$$\therefore V_o = -(4V_1 + V_2 - 0.1V_3) \quad \text{--- (1)}$$

Comparing (1) with $V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$, we get:-

$$\frac{R_f}{R_1} = 4 \quad \therefore \frac{R_f}{4} = R_1$$

$$\frac{R_f}{R_2} = 1 \quad \therefore R_2 = \frac{R_f}{1}$$

$$\frac{R_f}{R_3} = -0.1 \quad \therefore R_3 = -\frac{R_f}{0.1}$$

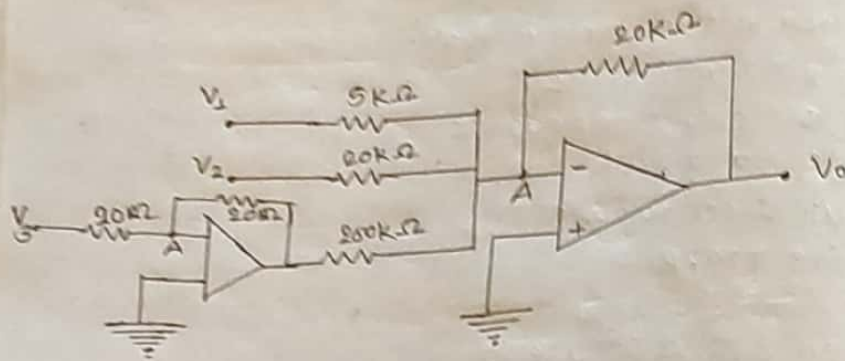
Let $R_f = 20 \text{ K}\Omega$.

$$\therefore R_1 = \frac{20}{4} = 5 \text{ K}\Omega$$

$$\therefore R_2 = \frac{20}{1} = 20 \text{ K}\Omega$$

$$\therefore R_3 = \frac{-20}{0.1} = -200 \text{ K}\Omega$$

Thus the required summing amplifier circuit that will produce an output voltage, $V_o = -4V_1 - V_2 + 0.1V_3$ is



6.a. What are the essential condition for oscillation?
Explain Wein bridge oscillator with circuit Diagram.
Derive the expression for the frequency of oscillator.
eg The closed loop gain of positive feedback amplifier is given by

$$A_f = \frac{V_o}{V_{in}} = \frac{A}{1-AB}$$

where, A = gain of an amplifier
 B = feedback ratio (factor)

when $AB = 1$

$$A_f = \frac{A}{1-1} = \frac{A}{0} = \infty$$

This means output may appear with very small input, ideally no input can also produce an output.
So the Barkhausen Criteria for oscillations are as follows:

- i) Loop gain must be unity i.e. $AB = 1$.
- ii) There must be positive feedback i.e. input and feedback should be in same phase.

Wein Bridge Oscillator:

In the given figure, the feedback signal V_f is connected to the non-inverting input terminal so the op amp works as an non-inverting amplifier.

Here,

$$Z_1 = R_1 - jX_C = R_1 - \frac{j}{\omega C_1}$$

$$Z_3 = R_3$$

$$Z_4 = R_4$$

$$X_C = \frac{1}{j\omega C_2} = -\frac{j}{\omega C_2}$$

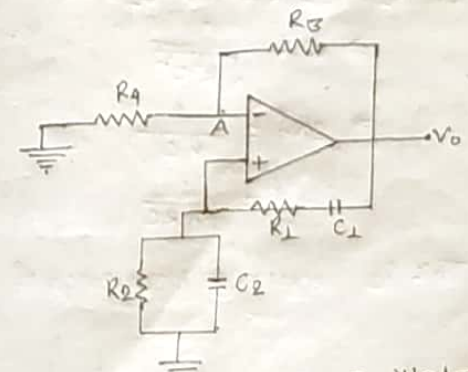


fig. Wein Bridge Oscillator

Neglecting the effect of op amp input and output impedance
At balance condition,
 $Z_1 \cdot Z_4 = Z_2 \cdot Z_3$

$$\Rightarrow \frac{(R_1 - \frac{j}{\omega C_1}) \cdot R_4}{Z_2} = R_3$$

$$\Rightarrow (R_1 - \frac{j}{\omega C_1}) \cdot Y_2 = \frac{R_3}{R_4}$$

$$\Rightarrow (R_1 - \frac{j}{\omega C_1}) \cdot (\frac{1}{R_2} + j\omega C_2) = \frac{R_3}{R_4}$$

$$\Rightarrow \frac{R_1}{R_2} + j\omega R_1 C_2 - \frac{j}{\omega R_2 C_1} + \frac{\omega C_2}{\omega C_1} = \frac{R_3}{R_4}$$

$$\Rightarrow \frac{R_1}{R_2} + \frac{C_2}{C_1} + j\omega R_1 C_2 = \frac{R_3}{R_4} + \frac{j}{\omega R_2 C_1}$$

Equating real and imaginary terms, we get:

$$\frac{R_1}{R_2} + \frac{C_2}{C_1} = \frac{R_3}{R_4} \quad \text{--- (1)}$$

Also, $\omega R_1 C_2 = \frac{1}{\omega R_2 C_1}$

$$\Rightarrow \omega^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

Let $R_1 = R_2 = R$ and $C_1 = C_2 = C$ Then,

$$\Rightarrow \omega = \frac{1}{\sqrt{R^2 C^2}}$$

$$\Rightarrow 2\pi f = \frac{1}{RC}$$

$$\therefore f = \frac{1}{2\pi RC}$$

This is required frequency of weing bridge oscillator.

(2018 - Spring)

6. a. Compare characteristics of ideal and practical op amp. Design an op amp circuit to produce output $V_o = 4V_1 + 5V_2 - 0.5V_3$, where V_1, V_2 and V_3 are inputs and V_o is output.

sol. Comparison of characteristics of ideal and practical op amp are as follows:

S.N.	Parameters	Ideal op amp	Practical op amp
1.	voltage gain	Infinite (∞)	Very high
2.	Input Impedance	Infinite (∞)	very high
3.	Output Impedance	zero (0)	very low
4.	Offset Voltage	zero (0)	Non zero
5.	Common Mode Rejection Ratio (CMRR)	Infinite (∞)	very high
6.	slew rate	Infinite (∞)	very high
7.	Bandwidth	Infinite (∞)	very high

we have,

$$V_o = 4V_1 + 5V_2 - 0.5V_3$$

$$\Rightarrow V_o = -(-4V_1 - 5V_2 + 0.5V_3) \quad \text{--- (2)}$$

Comparing (1) with $V_o = -(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3)$, we get

$$\frac{R_f}{R_1} = -4$$

$$\therefore R_1 = -\frac{R_f}{4}$$

$$\frac{R_f}{R_2} = -5$$

$$\therefore R_2 = -\frac{R_f}{5}$$

$$\frac{R_f}{R_3} = 0.5$$

$$\therefore R_3 = \frac{R_f}{0.5}$$

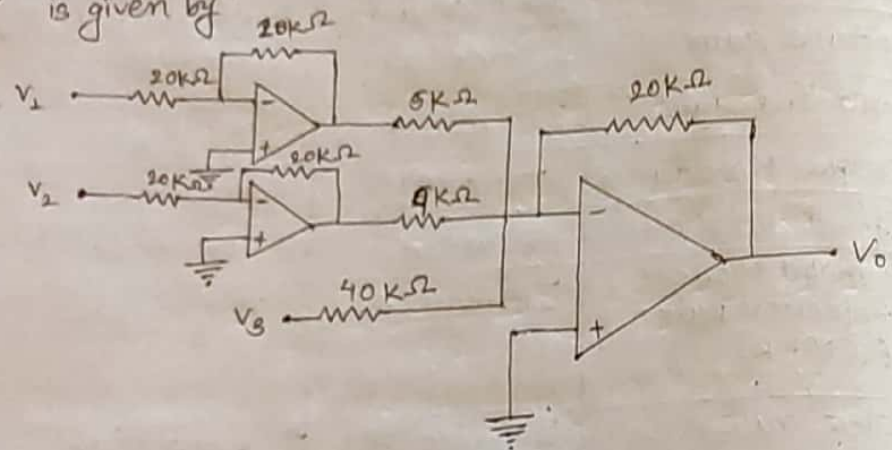
Suppose $R_f = 20K\Omega$

$$R_1 = \frac{-20}{4} = -5K\Omega$$

$$R_2 = \frac{-20}{5} = -4K\Omega$$

$$R_3 = \frac{20}{0.5} = 40K\Omega$$

Thus the required op amp circuit to produce output is given by



6.b) state Barkhausen's criteria for oscillation. Explain Wien bridge oscillator and compute its frequency of oscillation.

89) The closed loop gain of positive feedback amplifier is given by

$$A_f = \frac{V_o}{V_{in}} = \frac{A}{1-AB}$$

where, A = gain of an amplifier

B = feedback ratio (factor)

when $AB = 1$

$$A_f = \frac{A}{1-1} = \frac{A}{0} = \infty$$

This means output may appear with very small input, ideally no input can produce an output.

So the Barkhausen Criteria for oscillation are as follows:

- i) Loop gain must be unity i.e. $AB = 1$.
- ii) There must be positive feedback i.e. input and feedback must be in same phase.

Wien Bridge oscillator

Here, In the given figure, the feedback signal V_f is connected to the non-inverting input terminal so the op-amp works as an non-inverting amplifier. Here,

$$Z_1 = R_1 - jX_c = R_1 - \frac{j}{\omega C_1}$$

$$Z_3 = R_3$$

$$Z_4 = R_4$$

$$Y_2 = \frac{1}{R_2} + j\omega C_2$$

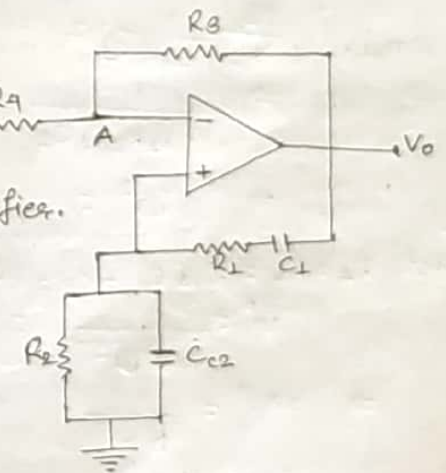


fig. Wien Bridge oscillator

Neglecting the effect of op amp input and output impedance

At balanced condition,

$$Z_1 \cdot Z_4 = Z_2 \cdot Z_3$$

$$\Rightarrow (R_1 - \frac{j}{\omega C_1}) R_4 = Z_2 R_3$$

$$\Rightarrow (R_1 - \frac{j}{\omega C_1}) R_4 = \frac{1}{Y_2} \cdot R_3$$

$$\Rightarrow Y_2 R_4 (R_1 - \frac{j}{\omega C_1}) = R_3$$

$$\Rightarrow (\frac{1}{R_2} + j\omega C_2) R_4 (R_1 - \frac{j}{\omega C_1}) = R_3$$

$$\Rightarrow R_4 (\frac{R_1}{R_2} - \frac{j}{\omega R_2 C_1} + j\omega R_1 C_2 + \frac{\omega C_2}{\omega C_1}) = R_3$$

$$\Rightarrow \frac{R_1}{R_2} - \frac{j}{\omega R_2 C_1} + j\omega R_1 C_2 + \frac{C_2}{C_1} = \frac{R_3}{R_4}$$

$$\therefore \frac{R_3}{R_4} + \frac{j}{\omega R_2 C_1} = \frac{R_1}{R_2} + \frac{C_2}{C_1} + j\omega R_1 C_2$$

On equating real and imaginary parts, we get;

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1} \quad (i)$$

And, $\frac{1}{\omega R_2 C_1} = \omega R_1 C_2$

$$\Rightarrow \frac{1}{R_1 R_2 C_1 C_2} = \omega^2$$

$$\Rightarrow \omega^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

Let $R_1 = R_2 = R$ & $C_1 = C_2 = C$. Then,

$$\omega^2 = \frac{1}{R^2 C^2} = \frac{1}{R^2 C^2}$$

$$\Rightarrow 2\pi f = \frac{1}{R C}$$

$$\therefore f = \frac{1}{2\pi R C}$$

This is the required frequency of oscillation.

(2017-Spring)

6. b. Design an adder circuit using op amp to obtain the output having expression as $V_o = -(3V_1 + 15V_2 + 120V_3)$.

We have,

$$V_o = -(3V_1 + 15V_2 + 120V_3) \quad \text{--- } (1)$$

Comparing (1) with $V_o = -(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3)$, we get;

$$\frac{R_f}{R_1} = 3$$

$$\therefore R_1 = \frac{R_f}{3}$$

$$\frac{R_f}{R_2} = 15$$

$$\therefore R_2 = \frac{R_f}{15}$$

$$\frac{R_f}{R_3} = 120$$

$$\therefore R_3 = \frac{R_f}{120}$$

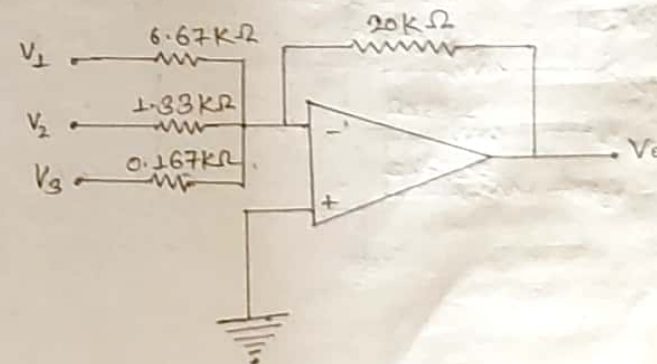
Let $R_f = 20 \text{ K}\Omega$

$$\therefore R_1 = \frac{20}{3} = 6.67 \text{ K}\Omega$$

$$\therefore R_2 = \frac{20}{15} = 1.33 \text{ K}\Omega$$

$$\therefore R_3 = \frac{20}{120} = 0.167 \text{ K}\Omega$$

Thus the required adder circuit using op amp with output voltage, $V_o = -(3V_1 + 15V_2 + 120V_3)$ is given by



6.b. Design an adder circuit using op amp to obtain the output having expression as $V_0 = 3V_1 - 15V_2 - 120V_3$.
 sol We have,

$$V_0 = 3V_1 - 15V_2 - 120V_3$$

$$\Rightarrow V_0 = -(3V_1 + 15V_2 + 120V_3) \quad \text{--- (1)}$$

Comparing (1) with $V_0 = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$, we get

$$\frac{R_f}{R_1} = -3 \quad \therefore R_1 = \frac{-R_f}{3}$$

$$\frac{R_f}{R_2} = 15 \quad \therefore R_2 = \frac{R_f}{15}$$

$$\frac{R_f}{R_3} = 120 \quad \therefore R_3 = \frac{R_f}{120}$$

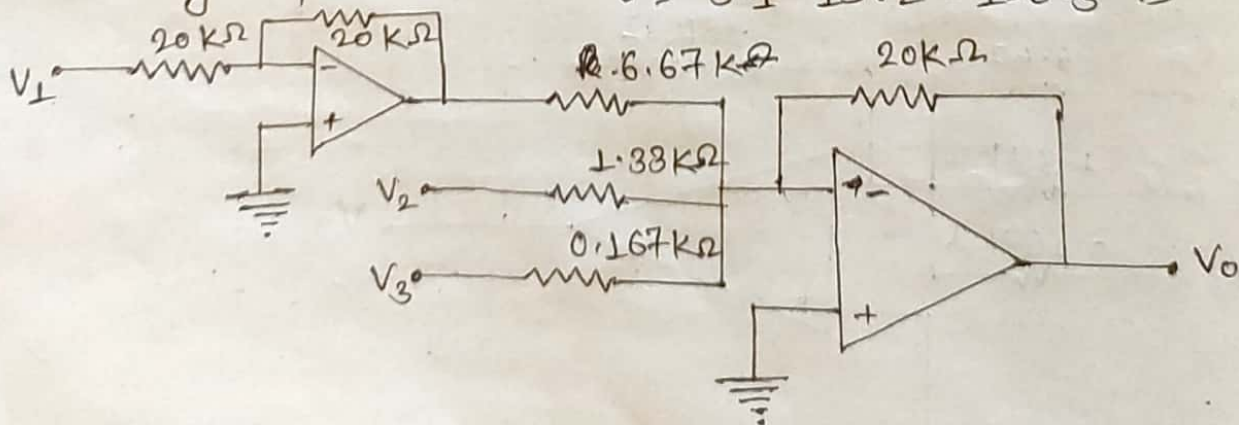
Let $R_f = 20 \text{ K}\Omega$. So

$$R_1 = \frac{-20}{3} = -6.67 \text{ K}\Omega = 6.67 \text{ K}\Omega$$

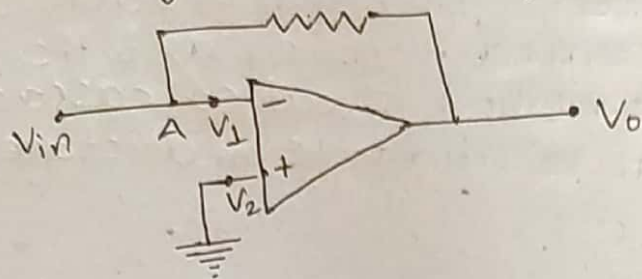
$$R_2 = \frac{20}{15} = 1.33 \text{ K}\Omega$$

$$R_3 = \frac{20}{120} = 0.167 \text{ K}\Omega$$

Thus the adder circuit using op amp to obtain the output having expression as $V_0 = 3V_1 - 15V_2 - 120V_3$ is



1. Virtual ground stability factor:
⇒



Due to the properties of operational amplifier with infinite gain, the voltage at inverting terminal reaches the voltage of non-inverting terminal. This property is called virtual ground or virtual short.

As we know,

$$V_o = A V_{id}$$

$$\Rightarrow V_o = A (V_2 - V_1)$$

For ideal op amp, $A = \infty$

$$\Rightarrow \frac{V_o}{V_2 - V_1} = A = \infty$$

$$\Rightarrow \frac{V_2 - V_1}{V_o} = \frac{1}{\infty}$$

$$\Rightarrow \frac{V_2 - V_1}{V_o} = 0$$

$$\Rightarrow V_2 - V_1 = 0$$

$$\therefore V_2 = V_1$$

This is called virtual short.

For $V_2 = 0V$,

$$V_1 = V_2 = 0V$$

This is called virtual ground.

2. Importance of cascading

⇒ The output of single stage amplifier is usually insufficient to drive an output device. So the output of each amplifier stage is coupled in some way to the input of next stage. This method is called cascading and the resulting system is said to be multistage amplifier.

A transistor containing more than one stage of amplification is known as multistage amplifier. In a multistage amplifier, a no. of single stage amplifiers are connected in cascade arrangement. That is the output of first stage is connected to the input of second stage through a suitable coupling device and so on.

Thus, to make the multistage amplifier, cascading is need.

3. Properties of non-linear device:

⇒ The diode which does not follow ohm's law ($V=IR$) or whose VI-characteristics curve is not a straight line is called a non-linear device.

The properties of non-linear devices are:-

- i) They do not follow ohm's law.
- ii) Their VI-characteristics is not a straight line.
- iii) They do not follow the network theorem like Thevenin's Theorem, Norton's Theorem, Superposition Theorem etc.
- iv) The circuits containing non-linear devices can be analyzed by using KCL or KVL.
- v) The internal resistance of non-linear devices depends on the operating conditions like temperature, current, voltage etc.
- vi) Examples: Diode, Transistor, FET etc.

4. Clipper Circuit:

⇒ A diode circuit which is used to clip-off or remove certain portion of input ac signal is called clipper circuit.

There are three types of clipper circuit:

i) Positive Clipper:

⇒ The clipper circuit which is used to remove certain positive portion of input ac signal is called positive clipper.



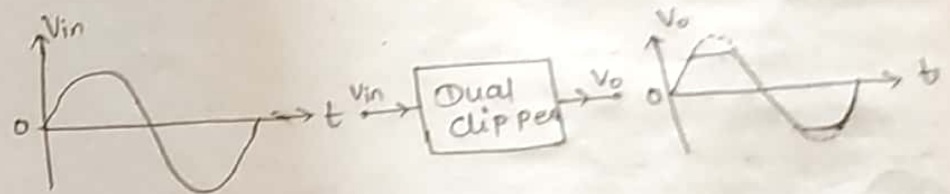
ii) Negative Clipper:

⇒ The clipper circuit which is used to remove certain negative portion of input ac signal is called negative clipper.



iii) Dual Clipper:

⇒ The clipper circuit which is used to remove certain ~~both portion~~ of both positive and negative portion of input ac signal is called dual clipper.



5. ~~Emitter follower~~ Emitter follower:
 ⇒ The voltage gain of common collector configuration is given by

$$A_v = \frac{R_E}{r_e + R_E}$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{R_E}{r_e + R_E}$$

Since, $r_e \ll R_E$, so,

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{R_E}{R_E}$$

$$\Rightarrow \frac{V_o}{V_{in}} = 1$$

$$\therefore V_o = V_{in}$$

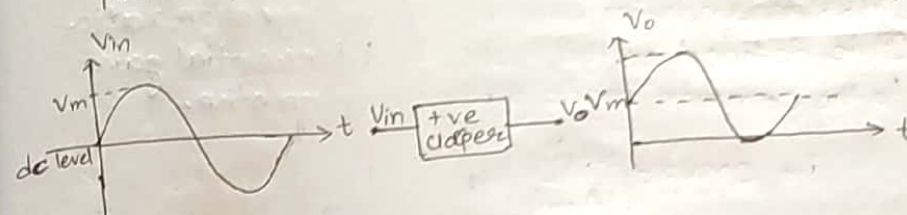
~~The~~ In common collector configuration, input voltage, V_{in} is supplied to base whereas the output voltage V_o is taken from emitter.

Thus $V_o = V_{in}$. So we can say that the output follows the input. Thus, common collector configuration is also called emitter follower.

6. Clamping Circuits
 ⇒ The diode circuit which is used to add a dc signal to an ac signal is called clamping circuit.
 There are two types of clamping circuits:

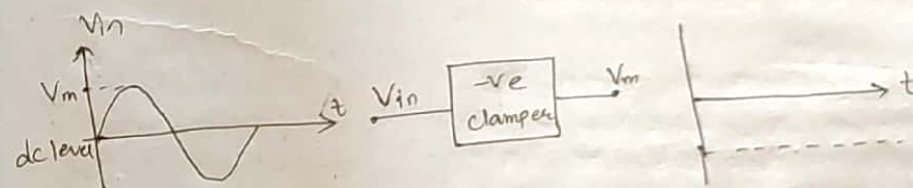
i) Positive clamping

⇒ The ~~diode~~ ^{clamping} circuit which is used to add positive dc voltage to an input ac signal is called positive clamping.



ii) Negative clamping

⇒ The ~~diode~~ ^{clamping} circuit which is used to add negative dc voltage to an input ac signal is called negative clamping.



7. ~~Signal~~ Oscillator: Slew rate and CMRR

Slew rate is defined as the maximum rate of change of output voltage of op-amp with respect to time.

$$\text{i.e. } SR = \left. \frac{dV_o}{dt} \right|_{\max}$$

CMRR stands for common mode rejection ratio. It is defined as the ratio of differential gain to the common mode gain of the op-amp.

$$\text{i.e. } CMRR = \frac{|A_d|}{|A_{cm}|}$$

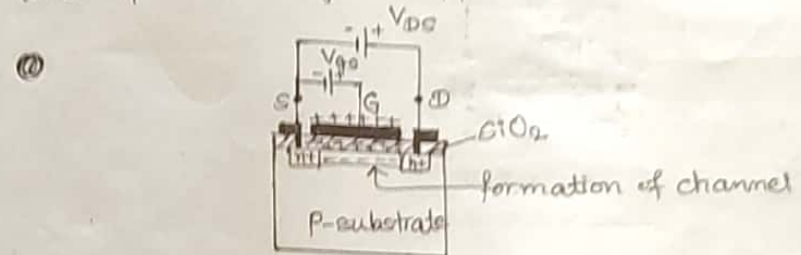
Ideally, $CMRR = \infty$. This means that the common signals like noises are perfectly rejected by the amplifier.

Enhancement MOSFET (EMOSFET) (Prabir)

construction

A N-channel EMOSFET is fabricated on p-type substrate. Two highly doped n⁺ region i.e. source and drain are created in the substrate. A thin layer of SiO₂ is grown on the substrate surface of substrate, covering the area between drain and source. Metal contacts are made in the source region, drain region, on the top of oxide layer and also on the substrate.

In EMOSFET, initially there is no channel between source and drain. Channels are electrically induced when positive V_{GS} is applied.



Operation:

case I: When $V_{GS} = 0$ and V_{DS} increasing

When $V_{GS} = 0$ and V_{DS} is increasing, there will be no drain current since there is no conducting path between drain and source.

With positive voltage of the gate, there will be the channel formed between source and drain. The minimum gate-to-source voltage at which significant current flow starts is called threshold voltage (V_{TR}).

case II: When $V_{GS} > V_{TR}$

When $V_{GS} > V_{TR}$, drain current increases rapidly and is given by

$$I_D = K (V_{GS} - V_{TR})^2$$

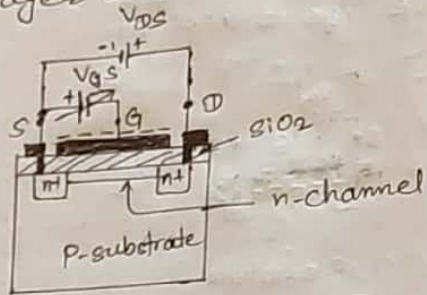
where $K = \text{constant}$

$V_{TR} = \text{Threshold voltage.}$

Depletion MOSFET (DMOSFET)

Construction:

A N-channel DMOSFET is fabricated on P-substrate. Two highly doped n⁺ region i.e. source and drain are created in the substrate. A thin layer of SiO₂ is grown on the surface of substrate, covering the area between source and drain. Metal contacts are made in the source region, drain region, on the top of layer and also on the substrate.



In DMOSFET ~~there~~ initially there exists a lightly doped n-channel between two ~~near~~ highly doped drain and source. Thus the current ~~to~~ (drain) I_D flows even at $V_{gs} = 0$ if V_{ds} is applied.

Operation:

case I: When $-V_{gs}$ and V_{ds} is applied.

When $-V_{gs}$ and V_{ds} is applied, the channel width reduces due to electrostatics induction i.e. the negative charge on Gate repels electrons from channel to the p-substrate. Thus the drain current reduces. This mode of operation is called depletion mode.

case II: When $+V_{gs}$ and V_{ds} is applied.

When $+V_{gs}$ and V_{ds} is applied, the charge on the Gate attracts electrons from channel to the p-substrate, thus the channel width increases. Thus the drain current increases. This mode of operation is called enhancement mode.

Thus DMOSFET operates in depletion mode as well as enhancement mode.

Q. Define Pinch-off voltage? Prove that the conductance of JFET is proportional to the square root of I_{DS} and I_{DSS} .

Pinch-off voltage is defined as the value of V_{DS} at which I_D becomes constant when $V_{GS} = 0$.

Transconductance is defined as the ratio of change in drain current I_D to change in Gate-source voltage V_{GS} at constant drain-source voltage (V_{DS}).

$$\text{i.e. transconductance, } g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} = \text{constant}} \quad (1)$$

we know,

$$I_{DS} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad (2)$$

Differentiating w.r.t. V_{GS} , we get

$$\frac{dI_{DS}}{dV_{GS}} = I_{DSS} \times 2 \left(1 - \frac{V_{GS}}{V_P} \right) \times \left(-\frac{1}{V_P} \right)$$

$$\Rightarrow \frac{dI_{DS}}{dV_{GS}} = -\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right)$$

$$\Rightarrow g_m = \frac{dI_{DS}}{dV_{GS}} = -\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right) \quad (\because V_{GS} = 0)$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \sqrt{\frac{I_{DS}}{I_{DSS}}} \quad (\because \text{from (2)})$$

$$g_m = g_{m0} \sqrt{\frac{I_{DS}}{I_{DSS}}} \quad (\because g_{m0} = \frac{2I_{DSS}}{|V_P|})$$

$$\therefore g_m = g_{m0} \sqrt{\frac{I_{DS}}{I_{DSS}}}$$

Thus the transconductance of JFET is proportional to the square root of product of I_{DS} and I_{DSS} .